Task 1:

Home Wireless Network Setup (Cisco Packet Tracer)

Objective: Create a basic Home Wi-Fi network topology in Cisco Packet Tracer consisting of one central Wireless Router and three PC endpoints, saving the project workspace as HomeWiFiNetwork.pkt.

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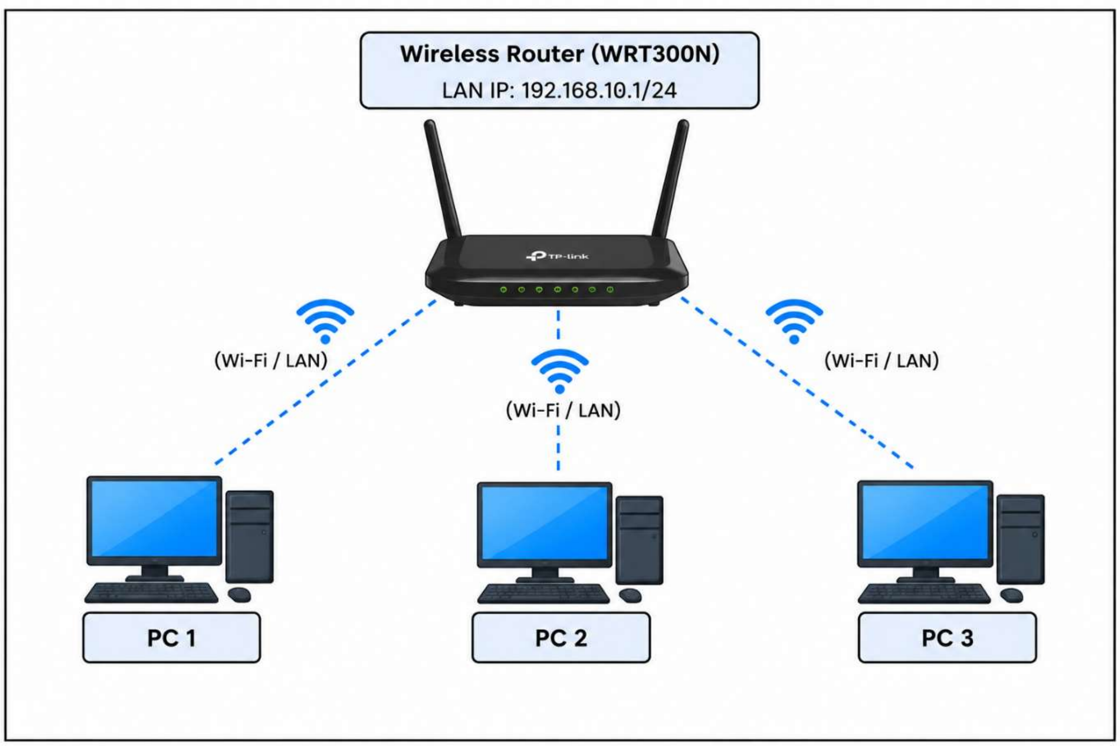
1. Network Topology Diagram:



2. Configuration & Setup Steps:

1.Initialize Project Workspace: Step 1.

Open Cisco Packet Tracer, create a new blank workspace, and click File > Save As... to save the project file as HomeWiFiNetwork.pkt.

2.Deploy Central Wireless Router: Step 2.

Navigate to Network Devices > Wireless Devices in the bottom toolbar and drag a Linksys WRT300N (or Wireless Router) onto the workspace grid.

3.Deploy Endpoint PCs: Step 3.

Navigate to End Devices and drag 3 PCs (PC0, PC1, PC2) into the workspace around the Wireless Router.

4.Connect PCs to Wireless Router: Step 4.

- Ethernet Connection Option: Select Connections > Copper Straight-Through and connect PC Ethernet ports to Ethernet Ports 1, 2, and 3 on the Wireless Router.
- Wireless Connection Option: Turn off each PC in physical view, swap the Ethernet module with a WMP300N Wireless Module, turn the PC back on, and connect to the default SSID of the wireless router.

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3. Verification:

To verify physical/wireless connectivity, check that all link lights between the three PCs and the Wireless Router show green status indicators, confirming active Layer 1 and Layer 2 link initialization.

Task 2:

IP Addressing Scheme & Static Configuration

Objective: Configure static IP addressing for all three PC endpoints within the 192.168.10.0/24 subnet and assign the default gateway pointing to the wireless router's internal interface IP (192.168.10.1).

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1. Network Addressing Table:

Device Name	Interface Port	Assigned IP Address	Subnet Mask	Default Gateway	Subnet Network ID
Wireless Router	LAN / Wireless Interface	192.168.10.1	255.255.255.0	*N/A (Gateway Node)*	192.168.10.0/24
PC 1	FastEthernet /	192.168.10.10	255.255.255.0	192.168.10.1	192.168.10.0/24

Device Name	Interface Port	Assigned IP Address	Subnet Mask	Default Gateway	Subnet Network ID
	Wireless				
PC 2	FastEthernet / Wireless	192.168.10.20	255.255.255.0	192.168.10.1	192.168.10.0/24
PC 3	FastEthernet / Wireless	192.168.10.30	255.255.255.0	192.168.10.1	192.168.10.0/24

2. Configuration Steps Executed in Packet Tracer:

1.Configure Wireless Router Gateway IP: Step 1.

Click on the Wireless Router, open the GUI tab > Setup > Basic Setup, set the IP Address to 192.168.10.1 and Subnet Mask to 255.255.255.0, then click Save Settings.

2.Configure Static IP on PC 1: Step 2.

Click on PC 1, open the Desktop tab, select IP Configuration, choose Static, and set:

- IP Address: 192.168.10.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

3.Configure Static IP on PC 2: Step 3.

Click on PC 2, navigate to Desktop > IP Configuration > Static, and set:

- IP Address: 192.168.10.20
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

4.Configure Static IP on PC 3: Step 4.

Click on PC 3, navigate to Desktop > IP Configuration > Static, and set:

- IP Address: 192.168.10.30
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

Task 3:

End-to-End Connectivity Testing & Verification

Objective: Validate internal Layer 3 network connectivity across all PC endpoints and the wireless gateway router by executing ICMP echo requests (ping tests) from the Cisco Packet Tracer Command Prompt.

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1. Ping Test Matrix & Verification Summary:

Source Device	Target Device	Target IP Address	Command Executed	Expected Packet Loss	Result
PC 1 (192.168.10.10)	Wireless Router	192.168.10.1	ping 192.168.10.1	**0% Loss**	**SUCCESS**
PC 1 (192.168.10.10)	PC 2	192.168.10.20	ping 192.168.10.20	**0% Loss**	**SUCCESS**

Source Device	Target Device	Target IP Address	Command Executed	Expected Packet Loss	Result
PC 1 (192.168.10.10)	PC 3	192.168.10.30	ping 192.168.10.30	**0% Loss**	**SUCCESS**
PC 2 (192.168.10.20)	PC 3	192.168.10.30	ping 192.168.10.30	**0% Loss**	**SUCCESS**

2. Execution Commands & Command Prompt Output:

Test 1: Ping from PC 1 to Gateway Wireless Router (192.168.10.1)

PC> ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=1ms TTL=128

Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

Test 2: Ping from PC 1 to PC 2 (192.168.10.20)

PC> ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.10.20: bytes=32 time=1ms TTL=128

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.20:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

3. Verification Conclusion:

- **Network Status:** All hosts respond with zero packet loss (0% Loss), confirming full bidirectional LAN connectivity and correct static IP and Default Gateway configuration across the entire network.

Task 4:

Comprehensive Network Documentation & Configuration Report

Project Title: Home Wi-Fi Network Setup & Verification

Simulation Tool: Cisco Packet Tracer

File Name: HomeWiFiNetwork.pkt

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1. Executive Summary & Topology Diagram:

This report documents the end-to-end design, static IP addressing, gateway setup, and ICMP connectivity testing for a small Home Wi-Fi Network created in Cisco Packet Tracer.

Network Topology Overview:

2. Complete IP Configuration & Device Specifications:

Device Name	Device Model	Interface	IP Address	Subnet Mask	Default Gateway	Connection Type
Wireless Router	Linksys WRT300N	LAN / Wireless	192.168.10.1	255.255.255.0	N/A	Central Gateway
PC 1	Generic PC	FastEthernet0	192.168.10.10	255.255.255.0	192.168.10.1	Copper / Wireless
PC 2	Generic PC	FastEthernet0	192.168.10.20	255.255.255.0	192.168.10.1	Copper / Wireless

Device Name	Device Model	Interface	IP Address	Subnet Mask	Default Gateway	Connection Type
PC 3	Generic PC	FastEthernet0	192.168.10.30	255.255.255.0	192.168.10.1	Copper / Wireless

3. Implementation & Configuration Steps:

1.Network Topology Deployment: Step 1.

Placed 1 Linksys WRT300N Wireless Router and 3 PCs onto the Cisco Packet Tracer logical workspace grid and saved the project as HomeWiFiNetwork.pkt.

2.Router Gateway Setup: Step 2.

Accessed the Wireless Router GUI dashboard under Setup > Basic Setup and set the local IP address to 192.168.10.1 with subnet mask 255.255.255.0.

3.Static IP Assignment on Endpoints: Step 3.

Configured each PC's network adapter with static IP parameters in the 192.168.10.0/24 subnet range, setting 192.168.10.1 as the primary default gateway for all devices.

4.Verification via Ping Diagnostics: Step 4.

Opened Command Prompt on PC 1 and executed ping 192.168.10.1 (Gateway), ping 192.168.10.20 (PC 2), and ping 192.168.10.30 (PC 3).

4. Verification Results:

Final Result: All ICMP ping packets returned 0% Packet Loss, confirming active Layer 1 (Physical), Layer 2 (Data Link), and Layer 3 (Network) functionality across the entire network topology.